

### Attempt #3

#### Modifier Equation for the $P(n)$ Equation

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My third attempt expanded upon the  $P(n)$  equation, shifting the focus away from the  $S_p(n)$  formula and into the  $P(n)$  equation. I postulated that, perchance, there was a multiplier equation that could multiply into the  $P(n)$  equation to adjust for the differences between  $P(n)$  and the actual prime at  $n$ .

By dividing the prime generated by  $P(n)$  with the actual  $n^{\text{th}}$  prime integer (between  $n = 10$  and  $n = 2000$ ), I get the following table:

| $n$ |  $P(n)$ |  $A_p$ |  $r$ |
|-----|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| 10  | 31                                                                                       | 29                                                                                      | $\frac{29}{31}$                                                                        |
| 20  | 74                                                                                       | 71                                                                                      | $\frac{71}{74}$                                                                        |
| 30  | 122                                                                                      | 113                                                                                     | $\frac{113}{122}$                                                                      |
| 40  | 174                                                                                      | 173                                                                                     | $\frac{173}{174}$                                                                      |
| 50  | 229                                                                                      | 229                                                                                     | $\frac{229}{229}$                                                                      |
| 60  | 285                                                                                      | 281                                                                                     | $\frac{281}{285}$                                                                      |
| 70  | 344                                                                                      | 349                                                                                     | $\frac{349}{344}$                                                                      |
| 80  | 404                                                                                      | 409                                                                                     | $\frac{409}{404}$                                                                      |
| 90  | 467                                                                                      | 463                                                                                     | $\frac{463}{467}$                                                                      |
| 100 | 531                                                                                      | 541                                                                                     | $\frac{541}{531}$                                                                      |

Generating a regression equation for the  $r$  column, I get the following:



Although the  $R^2$  value is not exactly equal to 1, it's still a proof of concept. I then put the equation into our  $P(n)$  equation in the following form:

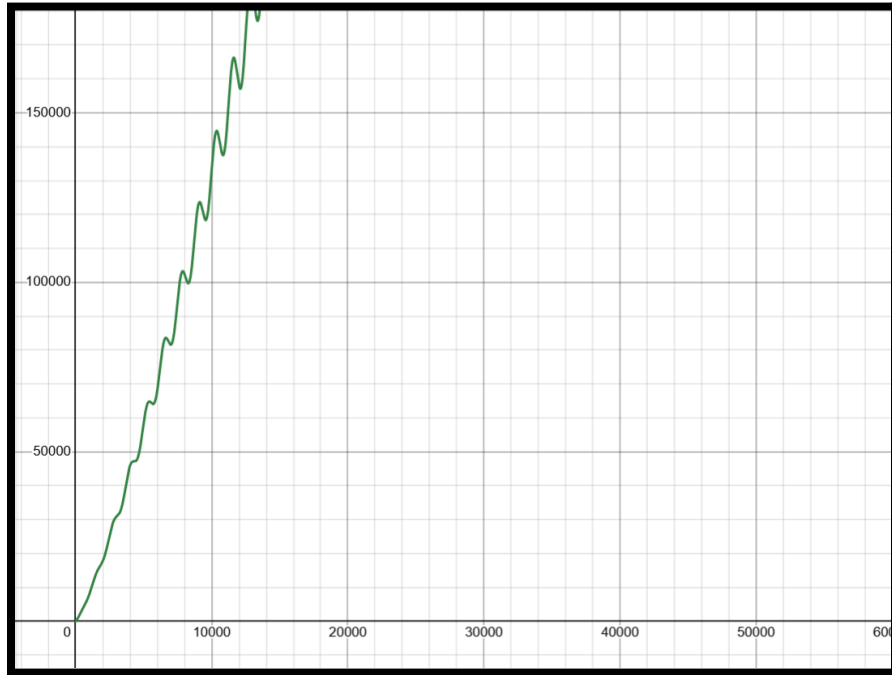
$$P(n) = \text{round} \left( f(n) \cdot \left( \frac{1}{S_p(n)} - \frac{1}{S_p(n-1)} \right) \right)$$

Where:

$$S_p(n) = \frac{1}{2 + \text{floor}(0.928914n^{2.20778})}$$

The rationale behind choosing the second  $S_p(n)$  equation generated rather than the first was because the second  $S_p(n)$  equation generated more accurate integers than the first  $S_p(n)$  equation.

The following graph was generated using the sinusoidal modifier equation to the original  $P(n)$  equation.



$$P(n) = (0.0538765 \cdot \sin(0.00497147n + 0.765209) + 0.948564) \cdot \left( \text{floor} \left( 0.928914n^{2.20778} \right) - \text{floor} \left( 0.928914(n-1)^{2.20778} \right) \right)$$

As seen below, the new  $P(n)$  equation generated is more accurate than the previous  $P(n)$  equation generated by Attempt #2.

| $n$  |  $P_{old}(n)$ |  $P_{new}(n)$ |  $A_p$ |
|------|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| 500  | 3725                                                                                             | 3511                                                                                             | 3571                                                                                        |
| 1000 | 8610                                                                                             | 7926                                                                                             | 7919                                                                                        |
| 1500 | 14054                                                                                            | 14037                                                                                            | 12553                                                                                       |
| 2000 | 19894                                                                                            | 17843                                                                                            | 17389                                                                                       |
| 2500 | 26048                                                                                            | 25532                                                                                            | 22307                                                                                       |
| 3000 | 32467                                                                                            | 30847                                                                                            | 27449                                                                                       |

Unfortunately, there is still a vast difference between the primes generated by the  $P(n)$  equation and the actual prime integers.



A new approach should be taken with Attempt #4, straying away from the proposed approach assumed in Attempt #2. Although the sinusoidal modifier equation assisted in the correction of the  $P(n)$  equation created in Attempt #2, it is clear from this attempt that the modifier was insufficient in correcting the primes generated by the  $P(n)$  equation. Perchance the creation of the  $n^{\text{th}}$  prime formula should be reimagined entirely.